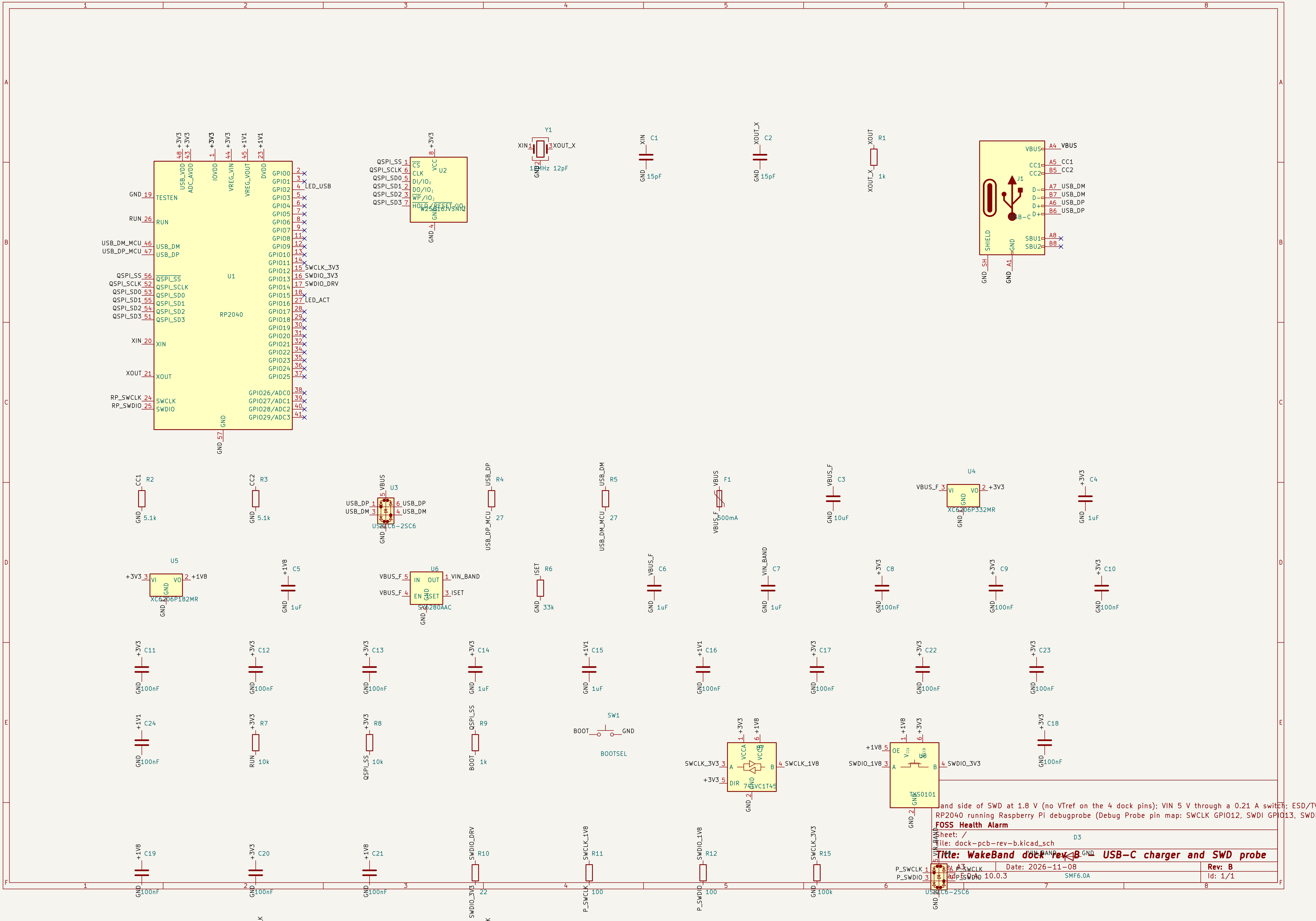
1
2
3
4
5
6
7
8
A
B
C
D
E
F
U1
RP2040
USB_VDD
ADC_AVDD
IOVDD
VREG_VIN
VREG_VOUT
DVDD
+3V3
+1V1
GND 19 TESTEN
RUN 26 RUN
USB_DM_MCU 46 USB_DM
USB_DP_MCU 47 USB_DP
QSPI_SS 56 QSPI_SS
QSPI_SCLK 52 QSPI_SCLK
QSPI_SD0 53 QSPI_SD0
QSPI_SD1 55 QSPI_SD1
QSPI_SD2 54 QSPI_SD2
QSPI_SD3 51 QSPI_SD3
XIN 20 XIN
XOUT 21 XOUT
RP_SWCLK 24 SWCLK
RP_SWDIO 25 SWDIO
GND 57
GPIO0 … GPIO25
GPIO26/ADC0
GPIO27/ADC1
GPIO28/ADC2
GPIO29/ADC3
4 LED_USB
15 SWCLK_3V3
16 SWDIO_3V3
17 SWDIO_DRV
27 LED_ACT
U2
W25Q16JVSNIQ
CS
CLK
DI/IO0
DO/IO1
WP/IO2
HOLD/RESET/IO3
VCC
GND
Y1
XIN
XOUT_X
12MHz 12pF
C1 XIN 15pF
C2 XOUT_X 15pF
R1 XOUT XOUT_X 1k
J1
USB-C
VBUS
CC1
CC2
D−
D+
SBU1
SBU2
SHIELD
A4 VBUS
A5 CC1
B5 CC2
A7 USB_DM
B7 USB_DM
A6 USB_DP
B6 USB_DP
A8
B8
GND SH
GND A1
R2 CC1 5.1k
R3 CC2 5.1k
U3
USBLC6-2SC6
USB_DP
USB_DM
VBUS
R4 USB_DP USB_DP_MCU 27
R5 USB_DM USB_DM_MCU 27
F1 VBUS VBUS_F 500mA
C3 VBUS_F 10uF
U4
XC6206P332MR
VBUS_F 3 VI
VO 2 +3V3
C4 +3V3 1uF
U5
XC6206P182MR
+3V3 3 VI
VO 2 +1V8
C5 +1V8 1uF
U6
SY6280AAC
VBUS_F 5 IN
OUT 1 VIN_BAND
VBUS_F 4 EN
ISET 3 ISET
R6 ISET 33k
C6 VBUS_F 1uF
C7 VIN_BAND 1uF
C8 +3V3 100nF
C9 +3V3 100nF
C10 +3V3 100nF
C11 +3V3 100nF
C12 +3V3 100nF
C13 +3V3 100nF
C14 +3V3 1uF
C15 +1V1 1uF
C16 +1V1 100nF
C17 +3V3 100nF
C22 +3V3 100nF
C23 +3V3 100nF
C24 +1V1 100nF
R7 +3V3 RUN 10k
R8 +3V3 QSPI_SS 10k
R9 QSPI_SS BOOT 1k
SW1
BOOTSEL
BOOT
GND
C18 +3V3 100nF
U7
74LVC1T45
VCCA
VCCB
SWCLK_3V3 3 A
B 4 SWCLK_1V8
+3V3 5 DIR
U8
TXS0101
OE
SWDIO_1V8 3 A
B 4 SWDIO_3V3
+1V8
C19 +1V8 100nF
C20 +3V3 100nF
C21 +1V8 100nF
R10 SWDIO_DRV SWDIO_3V3 22
R11 SWCLK_1V8 P_SWCLK 100
R12 SWDIO_1V8 P_SWDIO 100
R15 SWCLK_3V3 100k
D3
SMF6.0A
VIN_BAND
GND
U9
USBLC6-2SC6
P_SWCLK
P_SWDIO

band side of SWD at 1.8 V (no VTref on the 4 dock pins); VIN 5 V through a 0.21 A switch; ESD/TV
RP2040 running Raspberry Pi debugprobe (Debug Probe pin map: SWCLK GPIO12, SWDI GPIO13, SWD

FOSS Health Alarm

Sheet: /
File: dock–pcb–rev–b.kicad_sch

Title: WakeBand dock rev B – USB–C charger and SWD probe

Size: A3	Date: 2026–11–08	**Rev: B**
KiCad E.D.A. 10.0.3		Id: 1/1